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and dependencies among the variables may be too many. Then, it is difficult to build algorithms correctly. In such situations, we break an algorithm into parts, construct each part separately, and then integrate the parts to the complete algorithm.

The parts of an algorithm are known as functions. A function is like a sub algorithm. It takes an input, and produces an output, satisfying a desired input output relation.

Example 6.5. Suppose we want to calculate the surface area of a cylinder of radius r and height h.

$$A = 2\pi r^2 + 2\pi rh$$

We can identify two functions, one for calculating the area of a circle and the other for the circumference of the circle. If we abstract the two functions as circle_ area(r) and circle_circumference(r), then cylinder_area(r, h) can be solved as

cylinder_area (r,h) = 2 X circle_area (r) + circle_circumference (r) X h

6.4 Algorithm Design Techniques

There are a few basic principles and techniques for designing algorithms.

1. **Specification:** The first step in problem solving is to state the problem precisely. A problem is specified in terms of the input given and the output desired. The specification must also state the properties of the given input, and the relation between the input and the output.
2. **Abstraction:** A problem can involve a lot of details. Several of these details are unnecessary for solving the problem. Only a few details are essential. Ignoring or hiding unnecessary details and modeling an entity only by its essential properties is known as abstraction. For example, when we represent the state of a process, we select only the variables essential to the problem and ignore inessential details.
3. **Composition:** An algorithm is composed of assignment and control flow statements. A control flow statement tests a condition of the state and, depending on the value of the condition, decides the next statement to be executed.
4. **Decomposition:** We divide the main algorithm into functions. We construct each function independently of the main algorithm and other functions. Finally, we construct the main algorithm using the functions. When we use the functions, it is enough to know the specification of the function. It is not necessary to know how the function is implemented.

6.5 Specification

To solve a problem, first we must state the problem clearly and precisely. A problem is specified by the given input and the desired output. To design an algorithm for solving a problem, we should know the properties of the given input and the properties of the desired output. The goal of the algorithm is to establish the relation between the input and the desired output.